

Symbol	Unit	Meaning
<i>Indices</i>		
i, j	Token	Input (sold) and output (bought) endpoint tokens; the ordered pair is $i \rightarrow j$.
k	Token	Candidate vehicle token used as the route intermediate; $k \notin \{i, j\}$.
ℓ, p	Token / pool	ℓ indexes tokens in cross-token sums; p indexes liquidity pools.
t, w	UTC day / UTC week	t indexes calendar days; w indexes calendar weeks in settlement variables.
q	USD	Input quote notional. Un-suffixed route-cost columns use $q = \$10,000$.
r	Route unit	Receipt-audited coherent multihop route unit indexed by r .
<i>Route and liquidity aggregates</i>		
$A_t, \text{DirectRouteVolume}_t, \text{IndirectRouteVolume}_t$	USD	$\text{DirectRouteVolume}_t$ and $\text{IndirectRouteVolume}_t$ are day- t realized volumes over direct and indirect route units; A_t is their sum.
$B_{k,t}, B_t$	USD	$B_{k,t}$ is realized indirect-route volume through vehicle k ; $B_t = \text{IndirectRouteVolume}_t$ is total indirect-route volume on day t .
$N_{k,t}^B, N_t^B$	Route-unit count	$N_{k,t}^B$ counts indirect route units through k ; N_t^B counts all indirect route units on day t ; superscript B denotes bridged (indirect) routes.
$\mathcal{A}_t, \mathcal{A}_t^k, \mathcal{M}_t^k$	Sets of token pairs	$\mathcal{A}_t$ is the set of active endpoint pairs; $\mathcal{A}_t^k$ is the subset using k ; $\mathcal{M}_t^k$ is the subset for which k has the largest realized indirect-route volume.
$\text{Vol}_{k,t}^{\text{in}}, \text{Vol}_{k,t}^{\text{out}}$	USD	$\text{Vol}_{k,t}^{\text{in}}$ and $\text{Vol}_{k,t}^{\text{out}}$ are inbound and outbound route-leg volumes assigned to token k on day t .
$N_t^{\text{route}}, N_{k,t}^{\text{mid}}, N_{k,t}^{\text{src}}, N_{k,t}^{\text{sink}}$	Route count	N_t^{route} counts all day- t intent routes; $N_{k,t}^{\text{mid}}$, $N_{k,t}^{\text{src}}$, and $N_{k,t}^{\text{sink}}$ count routes assigning k the intermediate, source, and sink roles.
$\text{Vol}_t^{\text{route}}, \text{Vol}_{k,t}^{\text{mid}}, \text{Vol}_{k,t}^{\text{src}}, \text{Vol}_{k,t}^{\text{sink}}$	USD	$\text{Vol}_t^{\text{route}}$ is total day- t intent-route volume; $\text{Vol}_{k,t}^{\text{mid}}$, $\text{Vol}_{k,t}^{\text{src}}$, and $\text{Vol}_{k,t}^{\text{sink}}$ assign that volume to k by route role.
$L_{k,t}, \mathcal{K}$	USD / set of tokens	$L_{k,t}$ is vehicle-linked liquidity for k on day t ; $\mathcal{K}$ is the candidate-vehicle set.
$\mathcal{L}_{k,t}, \text{TVL}_{p,t}$	Set of pools / USD	$\mathcal{L}_{k,t}$ is the set of eligible pools linked to k on day t ; $\text{TVL}_{p,t}$ is total value locked in pool p on that day.
R_t^{WETH}	Daily log-return fraction	Day- t log return of the WETH price used to construct downside stress.
<i>Route-cost quote objects</i>		
$\mathcal{P}_{k,t,q}$	Set of token pairs	Ordered endpoint pairs eligible for k and quoted on day t at notional q .
$\mathcal{D}_{k,t,q}, \mathcal{V}_{k,t,q}, \mathcal{C}_{k,t,q}, \mathcal{T}_{k,t,q}, \mathcal{W}_{k,t,q}$	Sets of token pairs	Subsets of $\mathcal{P}_{k,t,q}$ with a direct route ($\mathcal{D}$), a route via k ($\mathcal{V}$), both routes ($\mathcal{C} = \mathcal{D} \cap \mathcal{V}$), a thin direct route ($\mathcal{T}$), or a vehicle-route cost advantage ($\mathcal{W}$).
$D_{i,j,q,t}, V_{i,j,k,q,t}, T_{i,j,q,t}$	Indicator (0/1)	Indicators for direct-route availability (D), route-through- k availability (V), and an executable direct route returning less than $0.9q$ (T).
$O_{i,j,q,t}^D, O_{i,j,k,q,t}^V$	USD	Quoted output values; superscripts D and V denote direct and vehicle routes.
$\Delta C_{i,j,k,q,t}$	Basis points	10,000 $(O_{i,j,k,q,t}^V - O_{i,j,q,t}^D) / O_{i,j,q,t}^D$ on $\mathcal{C}_{k,t,q}$.
<i>Settlement objects and operators</i>		
$\mathcal{R}_{k,w}$	Set of route units	Receipt-audited matched route units using vehicle k in UTC week w .
$\mathcal{R}_{k,w}^{\text{transfer}}$	Set of route units	Members of $\mathcal{R}_{k,w}$ whose receipt logs a transfer of vehicle k .
$ \mathcal{A} , \mathbf{1}\{\cdot\}$	Count / indicator (0/1)	Cardinality of set $\mathcal{A}$ and an indicator equal to one when its condition is true.
$\text{Betweenness}^{\text{vol}}, \Delta_7$	USD weighting / 7 days	Superscript vol denotes USD weighting; for any daily variable X_t , $\Delta_7 X_t = X_{t+7} - X_t$.

Variable	Formula	Unit	Data column	Definition
<i>Vehicle-use measures</i>				
VehicleShare $_{k,t}$	$\frac{B_{k,t}}{B_t}$	Fraction (0–1)	bridge_share	$B_{k,t}$ is indirect-route USD volume through k ; B_t is total indirect-route USD volume on day t .
AllRouteVehicleShare $_{k,t}$	$\frac{B_{k,t}}{A_t}$	Fraction (0–1)	all_route_bridge_share	$B_{k,t}$ is indirect-route USD volume through k ; A_t is total direct plus indirect route USD volume on day t .
VehicleCountShare $_{k,t}$	$\frac{N_{k,t}^B}{N_t^B}$	Fraction (0–1)	bridge_count_share	$N_{k,t}^B$ counts indirect route units through k ; N_t^B counts all indirect route units on day t .
PairCoverage $_{k,t}$	$\frac{ \mathcal{A}_t^k }{ \mathcal{A}_t }$	Fraction (0–1)	pair_coverage	$\mathcal{A}_t$ is the set of active endpoint pairs; $\mathcal{A}_t^k$ contains pairs using k in at least one indirect route on day t .
MainVehiclePairShare $_{k,t}$	$\frac{ \mathcal{M}_t^k }{ \mathcal{A}_t }$	Fraction (0–1)	pair_main_vehicle_share	$\mathcal{M}_t^k \subseteq \mathcal{A}_t$ contains pairs for which k has the largest realized indirect-route USD volume on day t .
VehicleVolume $_{k,t}$	$B_{k,t}$	USD	bridge_volume_usd	Sum of realized USD volume over indirect routes using vehicle k on day t .
<i>Network and route-denominator controls</i>				
VShare $_{k,t}$	$\frac{\text{Vol}_{k,t}^{\text{in}} + \text{Vol}_{k,t}^{\text{out}}}{\sum_{\ell} (\text{Vol}_{\ell,t}^{\text{in}} + \text{Vol}_{\ell,t}^{\text{out}})}$	Fraction (0–1)	vshare	$\text{Vol}_{k,t}^{\text{in}}$ and $\text{Vol}_{k,t}^{\text{out}}$ are inbound and outbound route-leg USD volumes for k ; ℓ indexes every token in the day- t network.
Betweenness $_{k,t}$	$\frac{N_{k,t}^{\text{mid}}}{N_t^{\text{route}} - N_{k,t}^{\text{src}} - N_{k,t}^{\text{sink}}}$	Fraction (0–1)	betweenness_centrality	Fraction of day- t intent routes on which k is intermediate, excluding routes where k is the source or sink. Superscripts identify each route role.
Betweenness $_{k,t}^{\text{vol}}$	$\frac{\text{Vol}_{k,t}^{\text{mid}}}{\text{Vol}_t^{\text{route}} - \text{Vol}_{k,t}^{\text{src}} - \text{Vol}_{k,t}^{\text{sink}}}$	Fraction (0–1)	volume_weighted_betweenness	USD-volume analogue of Betweenness $_{k,t}$; each Vol term is day- t route volume assigned to the indicated role.
AllRouteVolume $_t$	$\text{DirectRouteVolume}_t + \text{IndirectRouteVolume}_t$	USD	daily_all_route_volume_usd	Total realized USD volume across direct and indirect route units on day t .
IndirectRouteVolume $_t$	B_t	USD	daily_indirect_route_volume_usd	B_t is total realized USD volume across indirect route units on day t .
IndirectRouteShare $_t$	$\frac{B_t}{A_t}$	Fraction (0–1)	indirect_route_share	B_t is indirect-route USD volume and A_t is all-route USD volume on day t .
<i>Liquidity measures</i>				
VehicleLiquidity $_{k,t}$	$L_{k,t} = \sum_{p \in \mathcal{L}_{k,t}} \text{TVL}_{p,t}$	USD	vehicle_linked_liquidity_usd	$\mathcal{L}_{k,t}$ is the set of eligible Uniswap V3 pools linked to k ; p indexes pools after filtering subgraph TVL outliers.
LPConc $_{k,t}$	$\frac{L_{k,t}}{\sum_{\ell \in \mathcal{K}} L_{\ell,t}}$	Fraction (0–1)	lp_concentration	$\mathcal{K}$ is the candidate-vehicle set; $L_{k,t}$ is vehicle-linked liquidity for k on day t .
LogVehicleLiquidity $_{k,t}$	$\ln(1 + L_{k,t})$	Natural-log points	log_vehicle_linked_liquidity	Natural log of one plus $L_{k,t}$ measured in USD.
<i>Route-cost opportunity measures</i>				
DirectAvailable $_{k,t,q}$	$\frac{ \mathcal{D}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	direct_available_share	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $D_{i,j,q,t} = 1$. The un-suffixed data column fixes $q = \$10,000$.
VehicleAvailable $_{k,t,q}$	$\frac{ \mathcal{V}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	vehicle_available_share	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $V_{i,j,k,q,t} = 1$, so both legs through k are executable.
NoDirectVehicleAvailable $_{k,t,q}$	$\frac{ \mathcal{V}_{k,t,q} \setminus \mathcal{D}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	no_direct_vehicle_available_share	Fraction of pairs with $D_{i,j,q,t} = 0$ and $V_{i,j,k,q,t} = 1$: no executable direct route but an executable route through k .
DirectDepth $_{k,t,q}$	$\text{median}_{\mathcal{D}_{k,t,q}} \left(O_{i,j,q,t}^D / q \right)$	Output USD per input USD	direct_depth_median	Median direct-route output ratio $O_{i,j,q,t}^D / q$ over executable-direct pairs $(i, j) \in \mathcal{D}_{k,t,q}$.
RouteCostAdvantage $_{k,t,q}$	$\text{median}_{\mathcal{C}_{k,t,q}} \Delta C_{i,j,k,q,t}$	Basis points	route_cost_advantage_median_bps	Median $\Delta C_{i,j,k,q,t}$ over common-support pairs $(i, j) \in \mathcal{C}_{k,t,q}$; positive values favor the route through k .
VehicleBeatsDirect $_{k,t,q}$	$\frac{ \mathcal{W}_{k,t,q} }{ \mathcal{C}_{k,t,q} }$	Fraction (0–1)	vehicle_beats_direct_share	Fraction of $(i, j) \in \mathcal{C}_{k,t,q}$ for which $\Delta C_{i,j,k,q,t} > 0$, equivalently $O_{i,j,k,q,t}^V > O_{i,j,q,t}^D$.
ThinDirectShare $_{k,t,q}$	$\frac{ \mathcal{T}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	thin_direct_share	Fraction of pairs with $D_{i,j,q,t} = 1$ and $O_{i,j,q,t}^D / q < 0.9$.
<i>Stress and dynamic variables</i>				
Stress $_t$	$\max\{-R_t^{\text{WETH}}, 0\}$	Daily log-return fraction	stress_downside	R_t^{WETH} is the day- t WETH log return; only downside moves enter.
StressEvent $_t$	$\mathbf{1}\{\text{Stress}_t \geq 0.08\}$	Indicator (0/1)	stress_event_8pct	Equals one when Stress $_t$ is at least 0.08 on day t .
FutureVehicleShare $_{k,t}^{(7)}$	$\text{VehicleShare}_{k,t+7}$	Fraction (0–1)	future_bridge_share_t7	Vehicle share for token k seven calendar days after day t .
$\Delta_7 \text{VehicleShare}_{k,t}$	$\text{VehicleShare}_{k,t+7} - \text{VehicleShare}_{k,t}$	Fraction points	delta_bridge_share_t7	Seven-day forward change for vehicle k , measured from day t .
<i>V4 settlement implementation measures</i>				
TransferIncidence $_{k,w}$	$\frac{ \mathcal{R}_{k,w}^{\text{transfer}} }{ \mathcal{R}_{k,w} }$	Fraction (0–1)	settlement_transfer_incidence	Fraction of receipt-audited route units $r \in \mathcal{R}_{k,w}$ containing an ERC-20 Transfer log for intermediate vehicle k .
ReceiptCount $_{k,w}$	$ \mathcal{R}_{k,w} $	Route-unit count	settlement_receipt_count	Number of receipt-audited matched route units for vehicle k in UTC week w .